RT-MLI Experimental Notes

Wednesday July 16, 2025

- SD DR was closed with RT-MLI mounts but no Zotefoam filters installed
- All joints had thermal grease on them and vacuum joints had vacuum grease on them
- SD Cool Down was started

Friday July 18, 2025

- Liquid Nitrogen was added to the cold trap to start the DR
- SD temperatures were still not cold enough to start the DR and had instead leveled out
- It was decided to investigate the thermometers to make sure they were reading right, therefore we started to warm up the SD
- **Temperatures Without Turning on DR (No Blankoffs or RT-MLI)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	39.0069	5.3746	9.0868	10.5715

- **Thermometer Locations**
  - CH1 is located on the 40K Flange/Shell
  - CH2 Is located on 4K Flange/Shell
  - CH5 is located in Still Flange
  - CH6 is located on Mixing Chamber Flange

Monday July 21, 2025

- Opened up SD to inspect Thermometers, but all seemed to be set up correctly
- The likely reason for the SD being too warm is due to the optical gap in the 4K and 40K shells which let in 300K radiation directly hit the Mixing Chamber Flange
- I added blankoff plates to the respective optical gaps and closed up the SD for a second cooldown tomorrow morning

Tuesday July 22, 2025

- Second Cooldown was started, PT was turned on but the DR will be turned on tomorrow morning

Wednesday July 23, 2025

- **Temperatures Without Turning on DR (With Blankoffs)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	37.9884	3.28277	4.01338	3.773

- DR was turned On
- **Temperatures With DR On (With Blankoffs)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	38.6909	2.72990	0.91834	0.035399

- Data was collected therefore SD was warmed up again
- Next stage is to install the RT-MLI and KIDs
    - KIDs may take another week to arrive

Friday July 25, 2025
- SD Finished Warming Up

Tuesday July 29, 2025
- KIDs arrived
- SD Vacuum Softened and Ready to Take Apart
- SD Shells were taken off

Wednesday July 30, 2025
- Zotefoam was mounted onto the SD shell
    - 2'' of filters on 300K Stage
    - 2'' of filters on 40K Stage

Friday August 1, 2025
- SD was closed up
- Cooldown started

Monday August 4, 2025
- **Temperatures Without Turning on DR (2 and 2 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	38.7	3.3	4.0	3.9

- There is a suspected leak in the vacuum can, after several leak checks no specific locations were discovered
    - Potential Reasoning for leak was that there wasn't enough of a path for air to escape Zotefoam filter on 300K Stage
- SD Warmup was started

Thursday August 7, 2025
- SD was opened up, it took some time to pump up because of the needle valve

Monday August 18, 2025

- DR had turned on a couple times but was not reaching the temperature expected
- Plate was added in front of window to troubleshoot
- **Temperatures With DR on and Plate Covering Window (2 and 2 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	39.89	2.76	0.640	1.01

- DR is still not cold enough, more troubleshooting to come
    - Still Heater was turned on and we saw the temperature of the Mixing Chamber Flange go down but still not enough
- Realized that there shouldn't be an optical gap in the 4K stage and that there needs to be a filter over the 4K stage
- Warmup procedure began and design for 4K Filter and Ring Clamp was submitted to be machined

Wednesday August 20, 2025

- SD Finished Warming up, Needle Valve was started to pump up
- SD was opened up and the Optical Box was taken off
- **Zotefoam Thickness Measurements:**

Zotefoam Layers	300K (13 Layers)	40K (12 Layers)
Thickness (in)	1.9 ± 0.0312	1.8 ± 0.0312

Friday August 22, 2025

- 4K Filter and Clamp were machined and mounted to the SD
- SD was closed up and Cool Down procedure was started

Monday August 25, 2025

- DR began condensing
- **Temperatures With DR OFF and Plate Covering Window and 4k Filter (2 and 2 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	38.00	3.265	3.844	3.651

- **Temperatures With DR ON and Plate Covering Window and 4k Filter (2 and 2 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	39.1430	2.9014	1.1548	0.0386

- SD began warm up script

Wednesday August 27, 2025

- SD is fully warmed up and now pumping up with the needle valve
- SD was opened up
- **Zotefoam Thickness Measurements:**

Zotefoam Layers	300K (13 Layers)	40K (12 Layers)
Thickness (in)	1.92 ± 0.002	1.82 ± 0.002

- Foam stacks were taken of the SD until optical box became available again

Wednesday September 10, 2025

- Still waiting on the optical box, but will now perform a cooldown with the foam filters and using an eco-absorber to test the thermal performance
- Zotefoam thickness was measured and it seemed to return back to its original thickness

Friday September 12, 2025

- Pump down and cool down were started
- Filter stack added back into the SD
- Due to internet issues, needle valve was controlled manually
- Cooldown should be done by monday

Monday September 22, 2025

- **Temperatures With DR OFF (2 and 2 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	38.63	3.261	3.839	3.648

- Began Condensing the SD

Wednesday September, 24

- **Still Heater is a 0%, typically it is set to 20-25% to optimize the cooling power**
  - This is why our Mixing Chamber Temperatures are slightly higher than usual
- **Temperatures With DR ON and (2 and 2 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	39.25	2.768	0.758	0.0466

- Ecosorb was added underneath

- **Trial 1 with Ecosorb and Still Heater Set at 0%**

	300K Ecosorb	70K Ecosorb
CH1 Starting Temp. (K)	39.256	39.292
CH1 Final Temp. (K)	39.292	39.304
CH2 Starting Temp. (K)	2.769	2.771
CH2 Final Temp. (K)	2.771	2.766
CH6 Starting Temp. (mK)	46.6	46.6
CH6 Final Temp. (mK)	46.6	45.4

- **Trial 2 with Ecosorb and Still Heater Set at 0%**

	300K Ecosorb	70K Ecosorb
CH1 Starting Temp. (K)	39.304	39.265
CH1 Final Temp. (K)	39.265	39.227
CH2 Starting Temp. (K)	2.766	2.771
CH2 Final Temp. (K)	2.771	2.770
CH6 Starting Temp. (mK)	45.4	46.5
CH6 Final Temp. (mK)	46.5	45.6

- **Still Heater was Set to 20%**
    - CH5 Temperature: 1.164 K
- **Trial 3 with Ecosorb and Still Heater Set at 20%**

	300K Ecosorb	70K Ecosorb
CH1 Starting Temp. (K)	39.227	39.62
CH1 Final Temp. (K)	39.62	39.65
CH2 Starting Temp. (K)	2.770	2.836
CH2 Final Temp. (K)	2.836	2.834
CH6 Starting Temp. (mK)	45.6	37.5
CH6 Final Temp. (mK)	37.5	36.2

- **Trial 4 with Ecosorb and Still Heater Set at 20%**

	300K Ecosorb	80K Ecosorb
CH1 Starting Temp. (K)	39.65	39.65
CH1 Final Temp. (K)	39.65	39.65
CH2 Starting Temp. (K)	2.834	2.837
CH2 Final Temp. (K)	2.837	2.839
CH6 Starting Temp. (mK)	36.2	37.7
CH6 Final Temp. (mK)	37.7	36.8

Wednesday October 8th, 2025

- SD was pressurized using the vent instead of the needle valve
    - This may have caused the filters to be damaged
    - Filters at the 40K Stage were damaged and now need to be replaced
- **Zotefoam Thickness Measurements:**

Zotefoam Layers	300K (13 Layers)	40K (12 Layers)
Thickness (in)	1.91	1.51

- Optical Box was installed into the SD
- We will now test 1 in thick Zotefoam for both layers

Friday October 10th, 2025

- 1 in thick Zotefoam layers were added (7 layers each)
- **Zotefoam Thickness Measurements:**

Zotefoam Layers	300K (7 Layers)	40K (7 Layers)
Thickness (in)	1.00	1.00

- SD was closed up and cooldown was started over the weekend

Wednesday October 15th, 2025

- SD Was Fulling Pumped Down and PT was Started
- Issue with the Needle Valve caused delays

Thursday October 16th, 2025

- SD Was Fully Cooled Down
- **Temperatures With DR OFF (1 and 1 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	38.55	3.347	3.935	3.750

Friday October 17th, 2025

- **Temperatures With DR ON Still Heater Set to 0% (1 and 1 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	38.76	2.849	0.757	0.056

- **Temperatures With DR ON Still Heater Set to 20% (1 and 1 inches of Zotefoam)**:

Thermometers	CH1	CH2	CH5	CH6
Temperatures (K)	39.05	2.947	1.161	0.043

- RFSoC was connected to SD
    - We had trouble reading out from using the RFSoC, all outputs seemed to be noise
    - We realized that this was due to the optical box cables being installed incorrectly, therefore we would need to open up again to fix it

Thursday October 23rd, 2025

- **Still Heater was Set to 20%**
    - CH5 Temperature: 1.165 K
- **Trial 1 with Ecosorb and Still Heater Set at 20%**

	300K Ecosorb	80K Ecosorb
CH1 Starting Temp. (K)	39.871	39.522
CH1 Final Temp. (K)	39.955	39.871
CH2 Starting Temp. (K)	2.902	2.908
CH2 Final Temp. (K)	2.906	2.902
CH6 Starting Temp. (mK)	42.5	43.8
CH6 Final Temp. (mK)	43.6	42.5

- **Trial 2 with Ecosorb and Still Heater Set at 20%**

	300K Ecosorb	80K Ecosorb
CH1 Starting Temp. (K)	40.016	39.955
CH1 Final Temp. (K)	39.974	40.016
CH2 Starting Temp. (K)	2.903	2.906
CH2 Final Temp. (K)	2.904	2.903
CH6 Starting Temp. (mK)	42.5	43.6
CH6 Final Temp. (mK)	43.7	42.5

- **Still Heater was Set to 0%**
    - CH5 Temperature: 0.770 K

- **Trial 3 with Ecosorb and Still Heater Set at 0%**

	300K Ecosorb	80K Ecosorb
CH1 Starting Temp. (K)	39.512	39.526
CH1 Final Temp. (K)	39.532	39.512
CH2 Starting Temp. (K)	2.843	2.845
CH2 Final Temp. (K)	2.842	2.843
CH6 Starting Temp. (mK)	55.5	55.3
CH6 Final Temp. (mK)	56.7	55.5

- **Trial 4 with Ecosorb and Still Heater Set at 0%**

	300K Ecosorb	80K Ecosorb
CH1 Starting Temp. (K)	39.535	39.532
CH1 Final Temp. (K)	3.445	39.535
CH2 Starting Temp. (K)	2.840	2.842
CH2 Final Temp. (K)	2.840	2.840
CH6 Starting Temp. (mK)	55.4	56.7
CH6 Final Temp. (mK)	57.4	55.4

- SD began warming up over the weekend

Wednesday October 29, 2025

- SD was opened up to fix connections to the Optical Box
- **Zotefoam Thickness Measurements:**

Zotefoam Layers	300K (7 Layers)	40K (7 Layers)
Thickness (in)	0.99	0.98

- Will now switch the 40K layers to be only 0.5 in thick

Thursday October 30, 2025

- **Zotefoam Thickness Measurements:**

Zotefoam Layers	300K (7 Layers)	40K (4r Layers)
Thickness (in)	1.00	0.54

- Needle Valve was fixed, the arduino had to be replaced

Friday October 31, 2025

- Checked wire connections using the LAN, everything seems in order
- SD was closed and the pump down procedure was started

Tuesday November 4, 2025

- SD was not cooling down enough, we started to warm up to investigate further
    - Possibly could be a touch
    - Or the PT is not working

Friday November 7, 2025

- SD was opened up but no signs of a touch anywhere
- We decided to hold off and perhaps switch up the filter setup

Tuesday November 18, 2025

- Zotefoam filters were switched from High Density (HD30)